



CMPG325 NETWORKS PROJECT

Milestone 1



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Introduction

North-West Cold Chain Logistics, based in Vryburg, operates in an industry where the reliable movement, storage, and monitoring of temperature-sensitive goods are crucial. Unlike traditional logistics, cold-chain logistics involves transporting and storing products under controlled temperature conditions. Keeping the environmental parameters consistent throughout the supply chain is vital for maintaining product quality, minimizing spoilage, and meeting regulatory and operational standards. Modern logistics heavily rely on information and communication technologies to enhance supply chain visibility. Devices such as sensors and IoT gadgets offer real-time data on shipment location, temperature, humidity, and other environmental factors. This data helps logistics teams monitor operations, detect issues early, and prevent significant losses. Digital systems also facilitate inventory management, shipment tracking, route planning, and communication across departments (DHL, 2026). Consequently, a reliable, scalable network infrastructure is essential for cold-chain logistics. It should ensure dependable connectivity among users, devices, and services, enabling efficient information flow as the organization grows. For North-West Cold Chain Logistics, the network design must adhere to the project's specific needs, utilizing the 192.168.28.0/24 IP range. It should provide adequate connectivity, support expansion into additional areas, and enable secure management of network devices via SSH. The implementation and testing will be performed using Cisco Packet Tracer.

1. Client Requirements

1.1 Client Information

Item	Requirement
Organisation	North-West Cold Chain Logistics (Vryburg)
Industry	Logistics
Client ID	CLI-046
Project ID	CMPG325-2026-046
Assigned Network	192.168.28.0/24
Network Challenge	SSH – Secure Device Management Plane
Simulation Platform	Cisco Packet Tracer

1.2 Network Requirements

The network solution for North-West Cold Chain Logistics must satisfy the following requirements:

No.	Client Requirement	Design Implication
1	Provide appropriate network connectivity for the assigned logistics scenario.	The network should contain the necessary networking devices and connections to allow communication between the required nodes.
2	Provide the appropriate network services required for the assigned scenario.	The network design should include the services necessary for the proposed network environment.
3	Use the assigned addressing block 192.168.28.0/24.	The IP addressing plan should be developed from this network block.
4	Use an appropriate network topology and device arrangement.	The physical and logical topologies must be designed to meet the client's requirements.
5	Configure the necessary routers, switches, end devices and other required nodes.	The topology should identify the devices required to implement the proposed network.
6	Accommodate the possibility of an additional floor being occupied next financial year.	The network design must allow for future expansion.
7	Accommodate Change Request CR2: the client takes over an additional floor/area and requires network coverage there.	The proposed network must provide a way for the additional floor/area to receive network coverage.
8	Provide secure device management using SSH.	Network devices must later be configured, verified and demonstrated using SSH for secure device management.
9	Allow successful end-to-end connectivity and testing.	The final Packet Tracer network must allow relevant nodes to communicate successfully and provide evidence of testing.
10	Be implemented and simulated using Cisco Packet Tracer.	The proposed design must be practical to implement in Packet Tracer.

1.3 Key Design Constraints

- ❖ The assigned IP addressing block is 192.168.28.0/24.
- ❖ An additional floor may be occupied by the client in the next financial year.
- ❖ The client has specifically requested network coverage for the additional floor/area through Change Request CR2.
- ❖ SSH is the assigned networking challenge and must be configured, verified and demonstrated.
- ❖ The final network must be implemented and simulated in Cisco Packet Tracer.

1.4 Requirement Summary

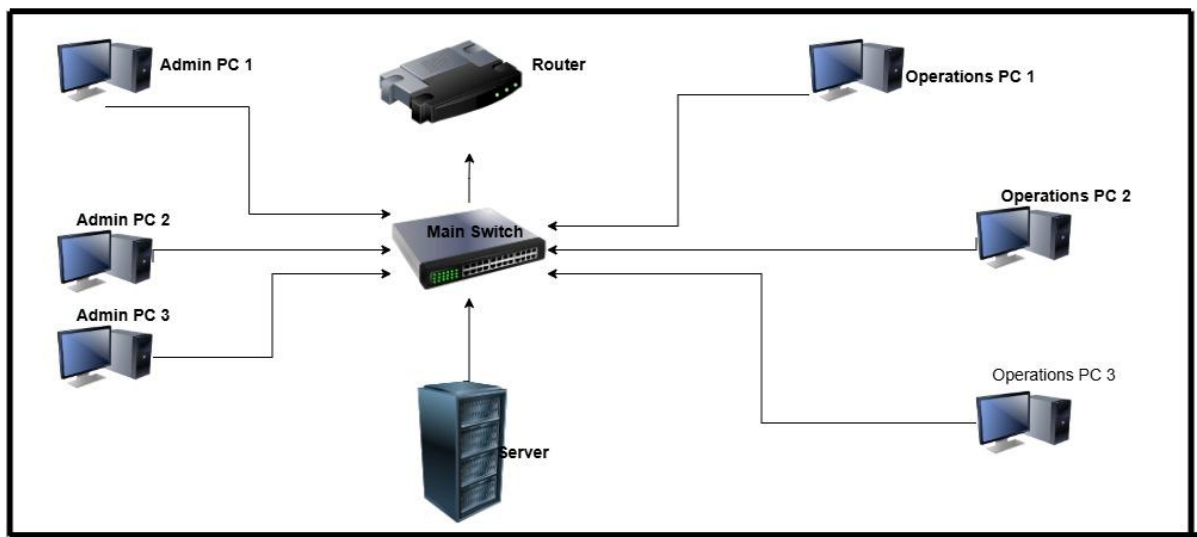
The proposed network must provide reliable connectivity for North-West Cold Chain Logistics while using the assigned 192.168.28.0/24 address space. The design must accommodate the client's future expansion into an additional floor/area and must include secure SSH-based management of network devices. The final design must be suitable for implementation, testing and demonstration in Cisco Packet Tracer.

Physical Topology

The physical topology was designed to represent a practical network environment for North-West Cold Chain Logistics. The current floor is divided into Administration, Operations, and a central Network Room, with the Network Room containing the main router, main switch and server. Three PCs are allocated to Administration and four PCs to Operations to represent the organisation's network users. A dedicated uplink from the main switch is provided for the future additional floor, where a second switch connects two representative user PCs and a wireless access point.

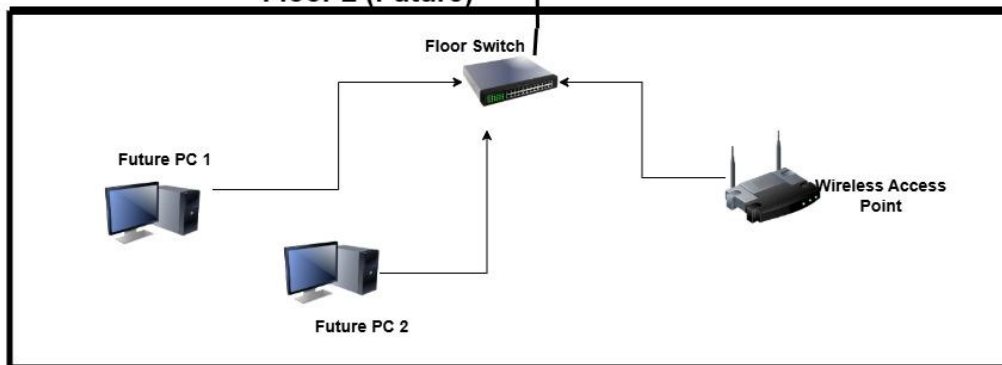
The project brief does not specify the exact number of employees, PCs, network devices, or physical layout of the organisation. Therefore, the number and placement of devices shown in the topology are reasonable design assumptions made to create a manageable and representative Packet Tracer network. The additional floor, its two PCs and wireless access point are also included as a representation of the client's future expansion requirement. These assumptions are intended to demonstrate scalability and connectivity while keeping the network practical to implement and test. The physical topology does not yet define IP addresses, VLANs or routing; these will be addressed in the logical topology and IP addressing plan. The requirement to accommodate the additional floor and use the assigned 192.168.28.0/24 network is based on the project brief.

Floor 1(Current)



UPLINK

Floor 2 (Future)

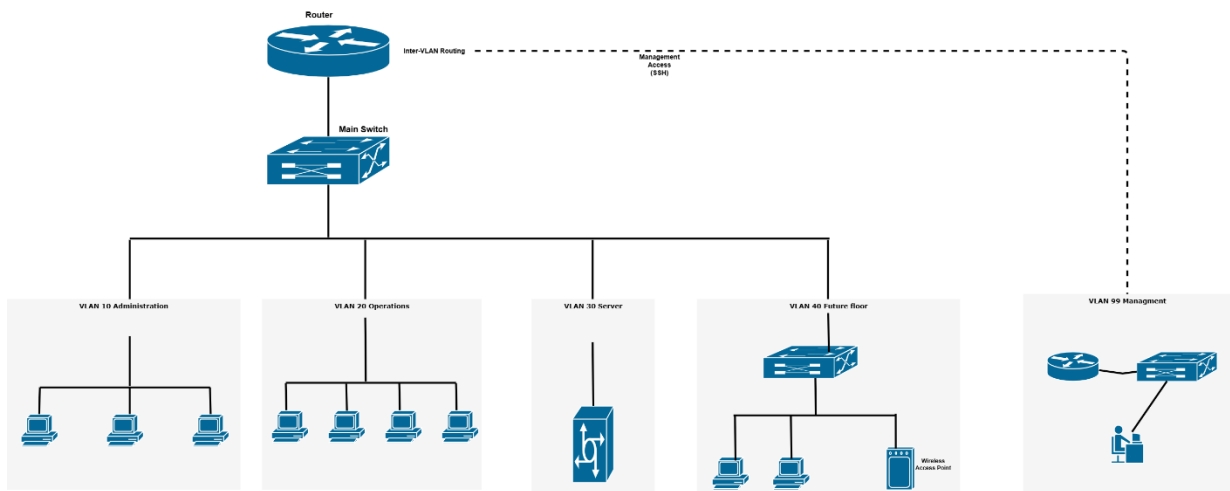


Logical Topology

The logical topology illustrates how the network is organised from a logical and communication perspective, showing the relationships between the different network segments, network devices and connected users. The design separates the organisation's network into functional areas to improve network organisation, traffic management and scalability. The router provides the necessary Layer 3 routing between the different network segments, while the switches provide connectivity to the users, server and future expansion area.

VLAN Structure

VLAN	Name	Purpose
10	Administration	Administration users
20	Operations	Operations users
30	Server	Network services
40	Future Floor	Future expansion
99	Management	Network-device management/SSH



IP Addressing Plan

VLAN	Network / CIDR	Subnet Mask	Usable Host Range	Gateway	Purpose
10	192.168.28.0/27	255.255.255.224	.1 – .30	.1	Administration
20	192.168.28.32/27	255.255.255.224	.33 – .62	.33	Operations
30	192.168.28.64/28	255.255.255.240	.65 – .78	.65	Server / Network Services
40	192.168.28.80/27	255.255.255.224	.81 – .110	.81	Future Floor
99	192.168.28.112/28	255.255.255.240	.113 – .126	.113	Network Management / SSH
—	192.168.28.128/25	255.255.255.128	.129 – .254	—	Reserved for Future Expansion

IP Addressing:

The **192.168.28.0/24** network assigned to the client was divided using Variable Length Subnet Masking (VLSM) to accommodate the five logical network segments identified in the proposed design. Larger **/27** subnets were allocated to the Administration, Operations and Future Floor networks to provide sufficient addresses for current users and future growth, while smaller **/28** subnets were allocated to the Server and Management networks. The first usable address in each subnet was reserved as the default gateway. Static addresses were assigned to network infrastructure and server devices, while end-user devices will obtain their addresses dynamically through DHCP. The remaining **192.168.28.128/25** address space has been reserved for future expansion, supporting the client's requirement to accommodate additional network infrastructure as the organisation grows.

Reference List

DHL Logistics of Things. (n.d.). *Home*. [online] Available at: <https://lot.dhl.com/> [Accessed 27 Aug. 2026].

Kurose, J.F. and Ross, K.W. (2013). *Computer Networking: a top-down approach*. Boston: Pearson.